

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

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About Me

I've been writing software for about 10 years, with the last three as a backend architect and technical lead. I use AI coding assistants and agentic workflows every day, with real opinions on where they help and where you need to understand the code yourself. My primary languages are TypeScript, Python, and Java/JVM, and I've worked with PostgreSQL throughout. At Arterys I built the Node.js and TypeScript services that ran medical AI algorithms on clinical imaging data. At AMA I built a self-service developer platform solo that took environment provisioning from 2-3 weeks to 2 hours (60x), and architected the backend for a pay-as-you-go product that has run at 99.9% uptime since launch. I like owning things end to end.

Work Experience

Alberta Motor Association

SOFTWARE DEVELOPER IV

Edmonton, Alberta

Jan 2023 - Present

- Built a self-service developer platform solo that took environment provisioning from 2-3 weeks down to 2 hours (60x). Self-service create/destroy, database snapshots, and pull-from-prod, adopted by 15 people across 3 teams and used to spin up 120+ environments. It made production debugging routine: a cancellation bug nobody could reproduce got root-caused in days using snapshotted production data. (Python, Terraform, AWS, Jenkins, Docker)
- Architected the infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts. 10 Lambda functions, 99.9% uptime since launch, 2.8M insured trips and 76,637 invoices processed. Won the 2024 Guidewire Innovation Award (1 of 3 worldwide). (Python, TypeScript, AWS Lambda, Terraform)
- Own 4 REST APIs serving 5 internal teams and external partners. Designed the contracts and versioning, approve new endpoint additions, and maintain OpenAPI design standards in Confluence. (Python, TypeScript, REST, OpenAPI)
- Set technical standards for a 12-person team through code reviews, RFC authorship, and PR checklists / API design guides. Authored an RFC moving the team to trunk-based development with feature flags. Worked in cross-functional pods alongside PM, design, and QA. Member of the Architecture Steering Committee, and mentor newer developers through legacy codebases and TDD. (Gosu/Java)
- Led a 2-3 engineer rewrite of a brittle legacy C# document generation system into containerized ASP.NET Core microservices (.NET 8). 4-5 services on a managed Kubernetes cluster, deployed with Helm. Wired up Datadog metrics for failure rate, throughput, generation time, and storage time. (C#, ASP.NET Core, Kubernetes, Docker, Helm, Datadog)
- Built an event-driven pipeline that ingests an external partner's policy data: S3 ingestion, a Kubernetes CronJob, a translation service, and transactional calls to a composite API, with OAuth 2.0 auth and Datadog monitoring. (Python, Kubernetes, AWS S3, OAuth 2.0)

Arterys Inc. (Acquired by Tempus Labs)

SOFTWARE DEVELOPER

Calgary, Alberta (Remote)

Aug 2021 - Aug 2022

- Built and maintained backend services that ran medical AI algorithms on clinical imaging data and returned results to the platform. (TypeScript, Node.js, AWS)
- Improved AI algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, catching and resolving issues before customer impact.

Alberta Motor Association

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Edmonton, Alberta

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2022

- Designed and built automated test framework that reduced defects by 30% on subsequent projects. Adopted across the insurance engineering team (15 developers / all Guidewire projects) and still in use 4+ years later. (Gosu/Java)
- Led 5-6 person development team on Guidewire insurance platform using TDD methodology. Performed Scrum Master role. Delivered projects on time and budget with no critical defects post-launch. Code reviews and mentoring junior developers.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local Guidewire development environments. Trained 5-6 developers on TDD methodologies through hosted training sessions.

Pleasant Solutions

SOFTWARE DEVELOPER

Edmonton, Alberta

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD methodology. Daily work in C# / .NET, heavily utilizing the MVC design pattern, with REST integration and tests written with NUnit and Moq. (C#, .NET, MVC, REST)

Lockheed Martin Canada

ASSOCIATE SOFTWARE ENGINEER

Halifax, Nova Scotia

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining data from multiple underwater and above-water sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system that enabled re-running failed test scenarios at will, improving debugging and testing capabilities.

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.
- Presented research through poster presentations.

Education

The Open University MSC. COMPUTING (SOFTWARE ENGINEERING) Currently pursuing Master's in Computing at The Open University.	Milton Keynes, England (Remote) May 2024 - Ongoing
The King's University B.S. IN COMPUTER SCIENCE	Edmonton, Alberta Sep 2012 - Dec 2016

Skills

AI-Assisted Development	I use Claude Code, Cursor, and GitHub Copilot every day, and I've formed real opinions on where they speed things up and where they quietly create work. I know when to trust the output and when to reason through the problem myself. I use these tools to move faster on things I already understand, not as a substitute for understanding them.
Software Craftsmanship	TDD has been core to how I work since 2018. I designed a test framework that reduced defects by 30% on subsequent projects and got adopted by 15 developers on the insurance engineering team; it's still in use four years later. I care about unit test coverage, continuous integration, and writing code that's easy to change. When a bug gets through, I want to understand why, fix it properly, and make sure it can't happen again.
Backend Services & Ownership	I've spent the last three years as the technical owner of a platform handling over \$300M in annual transactions, building APIs, infrastructure automation, and distributed systems. I own things end to end, from the design doc through the part where they have to stay up. I built a self-service developer platform from scratch that replaced a 2-3 week provisioning process with a 2-hour one, and I architected the backend for a pay-as-you-go product that has maintained 99.9% uptime since launch.
Reliability & Observability	I instrument what I ship with Datadog metrics for failure rate, throughput, and latency, and use them to set clear bars for what healthy looks like. I like chasing hard production failures to root cause: a cancellation bug nobody could reproduce got fixed in days once I could replay real production data against it. To me reliability is part of the feature, not a follow-up.

Technologies

Backend Languages	TypeScript, Node.js, Python, Java / Gosu (JVM), C# / .NET (ASP.NET Core, .NET 8), Bash
Architecture & APIs	RESTful APIs, OpenAPI, Microservices, Event-Driven Architecture, API Versioning
Cloud & Infrastructure	AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM, ECR), Terraform, Kubernetes, Docker, Azure DevOps, Azure Artifacts
Databases	PostgreSQL, SQL, Aurora/RDS, Data Modelling, Redis
AI-Assisted Development	Claude Code, Cursor, GitHub Copilot, Agentic Workflows
Observability	Datadog, Elasticsearch, Kibana, Metrics Design, SLOs, Alerting
CI/CD & Automation	CI/CD, GitHub Actions, Jenkins, GitOps, Feature Flags, Infrastructure as Code
Testing	NUnit, Moq, JUnit, TDD, Test Framework Design